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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCHE100 -

FUNDAMENTALS OF CHEMISTRY - II

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours
Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions, each question carries **1** mark

1. (d) No indicator is needed.
2. (b) Alkene
3. (c) The collision of gas molecules with the walls.
4. (c) To determine the concentration of an unknown substance.
5. (b) Formation of free radicals.
6. (d) All of the other answers.
7. (a) 0.5 M.
8. (d) The reaction rate depends on both the nucleophile and alkyl halide concentration.
9. (a) Cubic.
10. (b) 2-chloropropane. **[1x10 = 10]**

Part B

[Short Answer Questions, answer any **4** questions, each question carries **3** marks]

11. Explain the ionization process of acids and bases with relevant examples.

Acids

1. Strong Acids (e.g., HCl): Completely ionize in water, producing H^+ ions.

$HCl \rightarrow H^+ + Cl^-$

2. Weak Acids (e.g., CH_3COOH): Partially ionize in water, producing H+ ions.

$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$

Bases

1. Strong Bases (e.g., NaOH): Completely dissociate in water, producing OH- ions.

$NaOH \rightarrow Na^+ + OH^-$

2. Weak Bases (e.g., NH_3): Partially react with water, producing OH- ions.

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

(Explanation with relevant examples give full marks)

12. Give an account of electrophiles.

Electrophiles (meaning electron loving) are electron-deficient species that are attracted to electron-rich areas in molecules. They are characterized by an atom with an incomplete outer electron shell and therefore are very reactive. They are often positively charged species as exemplified by proton (H^+), cations such as (Cl^+, NO_2^+) or neutral species such as Lewis acids BF_3, $AlCl_3$.

(correct explanation with examples gives full marks)

13. What is meant by the term 'matter'?

Matter is anything that has mass and occupies space. It can exist in various forms, such as solid, liquid, or gas, and is composed of particles like atoms, molecules, or ions. Matter can undergo changes in state (solid, liquid, gas) and can be transformed through chemical reactions.

(correct explanation with any relevant three points give full marks)

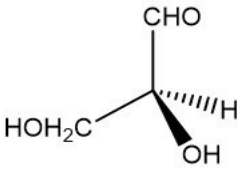
14. How does acid-base titration work? Provide an example to illustrate the process.

Acid-base titration is a method used to determine the concentration of an unknown acid or base by neutralizing it with a base or acid of known concentration. This process uses the principle of neutralization, where an acid reacts with a base to produce water and a salt. The indicator changes colour when the solution becomes neutral.

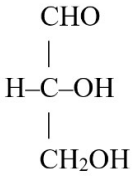
$HCl+NaOH \rightarrow NaCl+H_2O$

(Explanation-1.5 marks, Any one suitable example -1.5 marks)

15. Identify the following projection formula. Convert it into Fischer projection.



The given structure is a Haworth or wedge-dash projection of D-glyceraldehyde. The Fischer Projection formula is given below.



Orient the molecule so that the longest carbon chain is vertical. Place the most oxidized group (the aldehyde, CHO) at the top. Draw the chiral centre with horizontal bonds coming out of the plane and vertical bonds going into the plane. For the stereochemistry, the hydroxyl group (OH) on the chiral centre is on the right and the CH_2OH is on the bottom.

(Identification-1.5 marks, Fischer Projection formula -1.5 Marks)

16. What is compressibility factor (Z), and what does it indicate?

Z = PV / nRT, or Z= PVm/RT where P is pressure, V is volume, n is the number of moles, R is the gas constant, and T is temperature (Vm is the molar volume (V/n)).

- For ideal gases, Z = 1, indicating that the gas behaves according to the ideal gas law.
- For real gases, $Z \neq 1$, indicating deviations from ideal gas behavior due to intermolecular forces and molecular size.
- Z indicates how much a real gas deviates from ideal gas behavior.

(correct explanation give full marks) **[3x4 = 12]**

Part C

Short Essay Questions

Answer any **4** questions. Each question carries **7** marks

17. Explain the mole concept and Avogadro's number. How is the number of moles calculated from mass and volume? Give an example.

    The mole is defined as the amount of substance of a system that contains as many "elemental entities" (atoms, molecules, ions, electrons) as there are atoms in 12g of carbon-12 (i.e.6.023 × 10^{23}). This number 6.023 × 10^{23} is known as Avogadro number or Avogadro constant denoted by NA.

1. The mass of one mole of particles is called molar mass. The number of entities (atoms or molecules) contained in molar mass is equal to 6.023×10^{23}. Number of moles (n) = mass (m) / molar mass (M)

   Example: (1) One mole of sodium means 6.023 ×10^{23} Na atoms weighing 23.0 g (*GAM – Gram Atomic Mass* of sodium).

2. The volume occupied by one mole molecules of a gaseous substance is called Molar Volume or Gram Molecular Volume (GMV). One mole molecules of all gases occupy 22.4 litres at 273 K and 760 mm pressure (S.T.P). Hence, *molar volume of all gases at S.T.P is 22.4 litres.* Number of moles (n) = volume (V) / molar volume (22.4 L/mol at STP) Calculate the number of moles of oxygen gas (O2) occupying 44.8 liters at STP.

   Number of moles (n) = volume (V) / molar volume = 44.8 L / 22.4 L/mol = 2 moles

(Explanation of the mole concept and Avogadro's number-3 marks. Number of moles calculation from mass and volume with any one example -4 mark (2+2))

18. Define and explain the following terms with relevant examples (a) Polymer (b) Monomer (c) Polymerisation.

(a) ***Polymer***

A polymer is a large molecule composed of many repeated subunits, called monomers. Polymers can be found naturally, like cellulose, or synthesized, like plastics. Example: Polythene (polyethylene) is a polymer made up of many ethylene monomers.

(b) ***Monomer***

A monomer is a small molecule that reacts with other monomers to form a polymer. Monomers are the building blocks of polymers. Example: Ethylene (C_2H_4) is a monomer that polymerizes to form polythene.

(c) ***Polymerization***

Polymerization is the process of combining monomers to form a polymer. This can occur through various reaction mechanisms, such as addition or condensation reactions. Example: The polymerization of ethylene monomers to form polythene occurs through an addition reaction.

(Explanation with example: Polymer-2.5 marks, Monomer-2.5 mark, polymerisation-2 marks))

19. Explain the general properties of liquids and how they differ from solids and gases.

Properties of Liquids:

1. No definite shape: Takes the shape of the container
2. Definite volume: Maintains its volume
3. Fluidity: Can flow and take the shape of the container
4. Incompressibility: Relatively resistant to compression
5. Surface Tension: Exhibits surface tension, affecting its behavior
6. Viscosity: Has resistance to flow
7. Particle movement: Particles are free to move over each other, but are still attracted to each other.

Comparison to Solids and Gases:

- Solids: Have a fixed shape and volume, and are more rigid than liquids, do not flow
- Gases: Have neither a fixed shape nor a fixed volume, and are highly compressible, Particles are widely spaced and free to move randomly.

(Any 5 Properties- 5 marks, Compare it with solids and gases-2 marks)

20. Classify the substances as acids or bases according to Arrhenius, Bronsted-Lowry, and Lewis definitions, and explain why. (a) Ammonia (NH_3) reacts with water (H_2O) to form ammonium ion (NH_4^+) and hydroxide ion (OH^-). (b) Sulfuric acid (H_2SO_4) dissociates in water to produce H^+ and HSO_4^- ions. (c) Boron trifluoride (BF_3) reacts with ammonia (NH_3) to form an adduct.

Arrhenius Definition

- Acids: Substances that releases H^+ ion in water (e.g., HCl, HNO_3)
- Bases: Substances that releases OH^- ion in water (e.g., NaOH, KOH)

Bronsted-Lowry Definition

- Acids: Proton (H^+) donors (e.g., HCl, CH_3COOH)
- Bases: Proton (H^+) acceptors (e.g., NH_3, OH^-)

Lewis Definition

- Acids: Electron pair acceptors (e.g., BF_3, $AlCl_3$)
- Bases: Electron pair donors (e.g., NH_3, OH-)

(a) Ammonia (NH_3) Reaction

$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

- Ammonia accepts a proton (H^+) from water, forming ammonium ion (NH_4^+) and hydroxide ion (OH^-). This reaction shows ammonia's basic nature.

(b) Sulfuric Acid (H_2SO_4) Dissociation

$H_2SO_4 \rightarrow H^+ + HSO_4^-$

- Sulfuric acid donates a proton (H^+) to water, forming H^+ and HSO_4^- ions. This reaction demonstrates sulfuric acid's acidic properties.

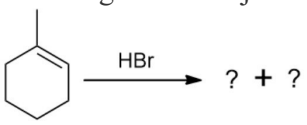
(c) Boron Trifluoride (BF_3) Reaction

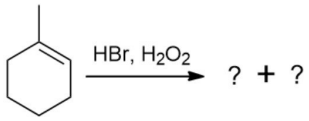
$BF_3 + NH_3 \rightarrow BF_3NH_3$ (adduct)

- Boron trifluoride (BF_3) accepts an electron pair from ammonia (NH_3), forming a coordinate covalent bond. This reaction illustrates BF_3's Lewis's acid behaviour, as it accepts an electron pair.

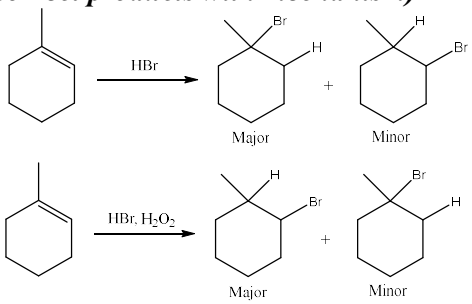
(Arrhenius, Bronsted-Lowry, and Lewis's definitions-3 marks Explanation to (a), (b) and (c) give 4 marks)

21. For each of the following reactions: i) Identify the major and minor products. ii) Explain your reasoning for the major/minor product formation. iii)Provide reaction mechanisms.

(a) 

(b) 

(a) The products formed according to Markownikoff rule. When an unsymmetrical addendum adds to an unsymmetrical alkene, the negative part of the addendum combines with the alkene carbon carrying the fewer number of hydrogen atoms. In the case of the halogen acids, the halogen atom is the negative part. The reaction follows carbocation intermediate. in the addition of hydrogen bromide to alkenes, the carbocation is first formed by the addition of proton and the carbocation then combines with bromide ion to form the final product. ***(3.5 mark for correct products with mechanism)***



(b) The products formed according to anti Markownikoff rule. the addition of hydrogen bromide to take place in the direction opposite to that predicted by Markownikoff rule. The reaction proceeds through free radical intermediates. ***(3.5 mark for correct products with mechanism)***

22. Differentiate between crystalline and amorphous solids.

Crystalline Solids:	**Amorphous Solids:**
1. Ordered Structure 2. Sharp Melting Point 3. Anisotropic Properties: Display different physical properties in different directions. 4. Definite Geometry 5. X-ray Diffraction: Produce sharp X-ray diffraction patterns, indicating a regular arrangement of particles. Eg. NaCl (rock salt), quartz, diamond	1. Lack a regular, three-dimensional arrangement of particles. 2. No sharp melting point. 3. Isotropic Properties: Display similar physical properties in all directions. 4. Irregular Shape 5. No X-ray Diffraction Eg. Glass, rubber, plastics

(Any 5 points with suitable examples – 7 marks) **[7x4 = 28]**